

SAPIENZA
UNIVERSITÀ DI ROMA

Integer Linear Programming

MIP problem catalogue

The integer and mixed integer linear programming problems used in the examples and exercises of the whole module, cited by their code ($P0$, $P1$, ...): problems with two, three and more variables, grouped by type. For each, the problem and the figure of its feasible region.

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Online version and downloadable material:
fabiofurini.github.io/integer-linear-programming/
github.com/fabiofurini/integer-linear-programming

Academic year 2026–2027

Contents

1 ILP problems with two variables – Type A **3**

1.1 ILP problem $P0$ . . . 4
1.2 ILP problem $P1$ . . . 5
1.3 ILP problem $P2$ . . . 6
1.4 ILP problem $P3$ . . . 7
1.5 ILP problem $P4$ . . . 8
1.6 ILP problem $P5$ . . . 9
1.7 ILP problem $P6$ . . . 10
1.8 ILP problem $P7$ . . . 11
1.9 ILP problem $P8$ . . . 12
1.10 ILP problem $P9$ . . . 13
1.11 ILP problem $P10$ . . . 14

2 ILP problems with two variables – Type B **15**

2.1 ILP problem $P11$ . . . 16
2.2 ILP Problem $P12$ . . . 17
2.3 ILP problem $P13$ . . . 18
2.4 ILP Problem $P14$ . . . 19

3 ILP problems with two variables – Type C **20**

3.1 ILP problem $P21$ . . . 21
3.2 ILP problem $P22$ . . . 22
3.3 ILP problem $P23$ . . . 23
3.4 ILP problem $P24$ . . . 24

4 ILP problems with three variables – Type A **25**

4.1 ILP problem $P41$ . . . 26
4.2 ILP problem $P42$ . . . 27
4.3 ILP problem $P43$ . . . 28
4.4 ILP problem $P44$ . . . 29
4.5 ILP problem $P45$ . . . 30
4.6 ILP Problem $P46$ . . . 31

5 ILP problems with three variables – Type B **32**
5.1 ILP problem $P51$ 33

6 ILP problems with three variables – Type C **34**
6.1 ILP problem $P61$ 35

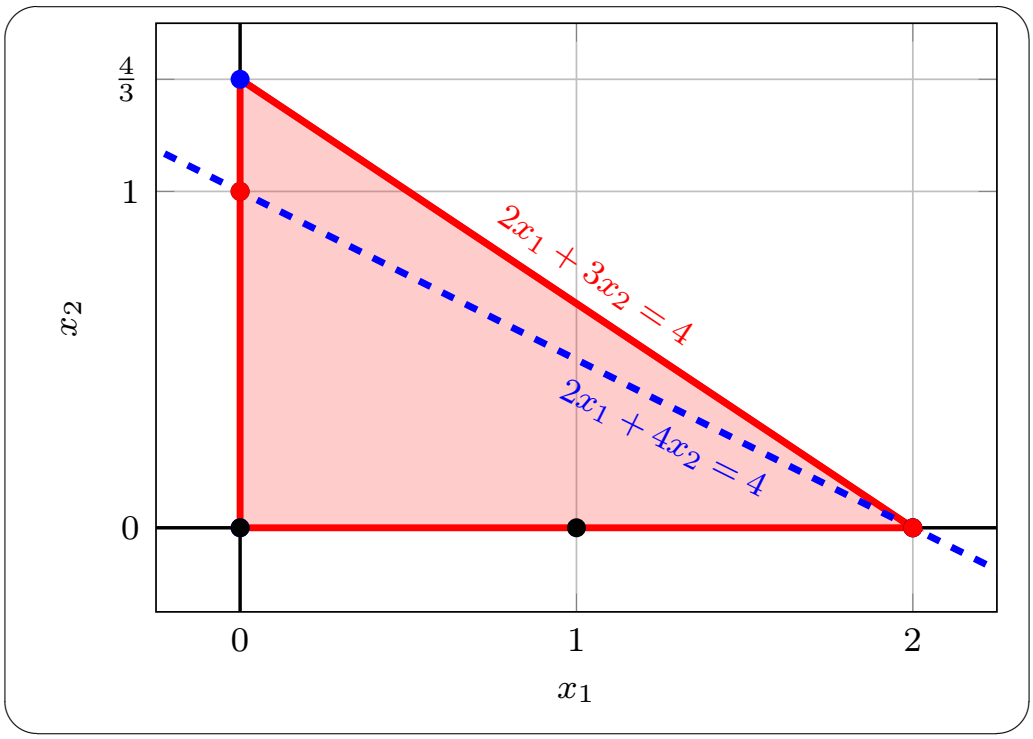
7 ILP problems with several variables **36**
7.1 ILP problem $P71$ 37
7.2 ILP problem $P72$ 38

Chapter 1

ILP problems with two variables – Type A

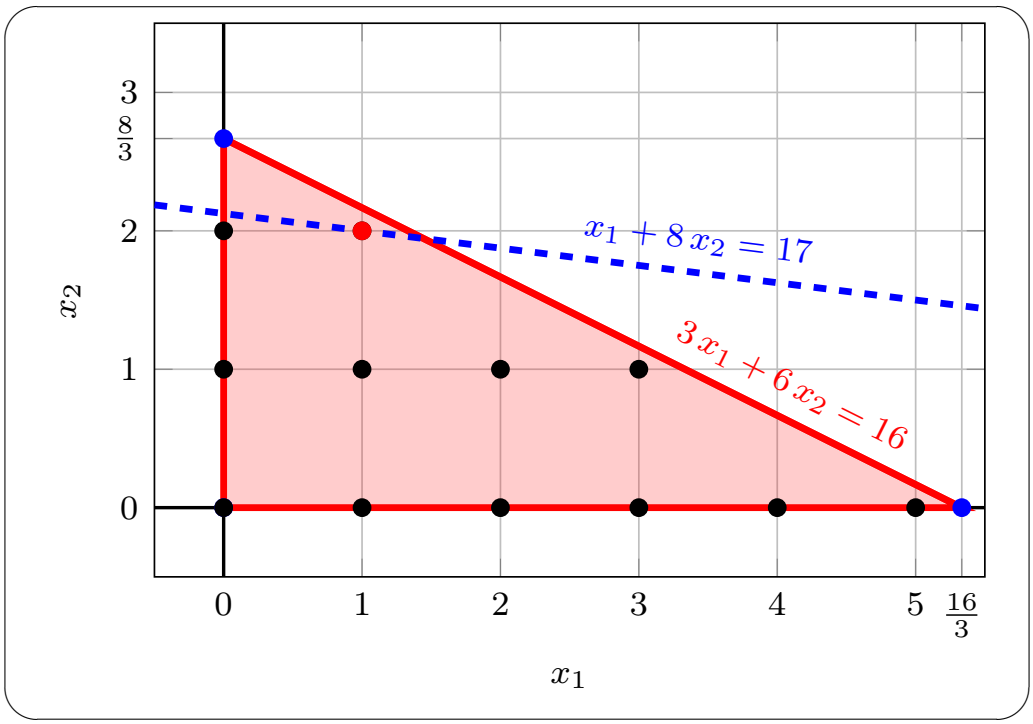
1.1 ILP problem *P0*

$$
\begin{array}{lrcrcrcl}
\max & 2 & x_1 & + & 4 & x_2 & & \\
\text{subject to} & 2 & x_1 & + & 3 & x_2 & \leq & 4 \\
& & x_1, & & & x_2 & \in & \mathbb{Z}_{\geq 0}
\end{array}
$$



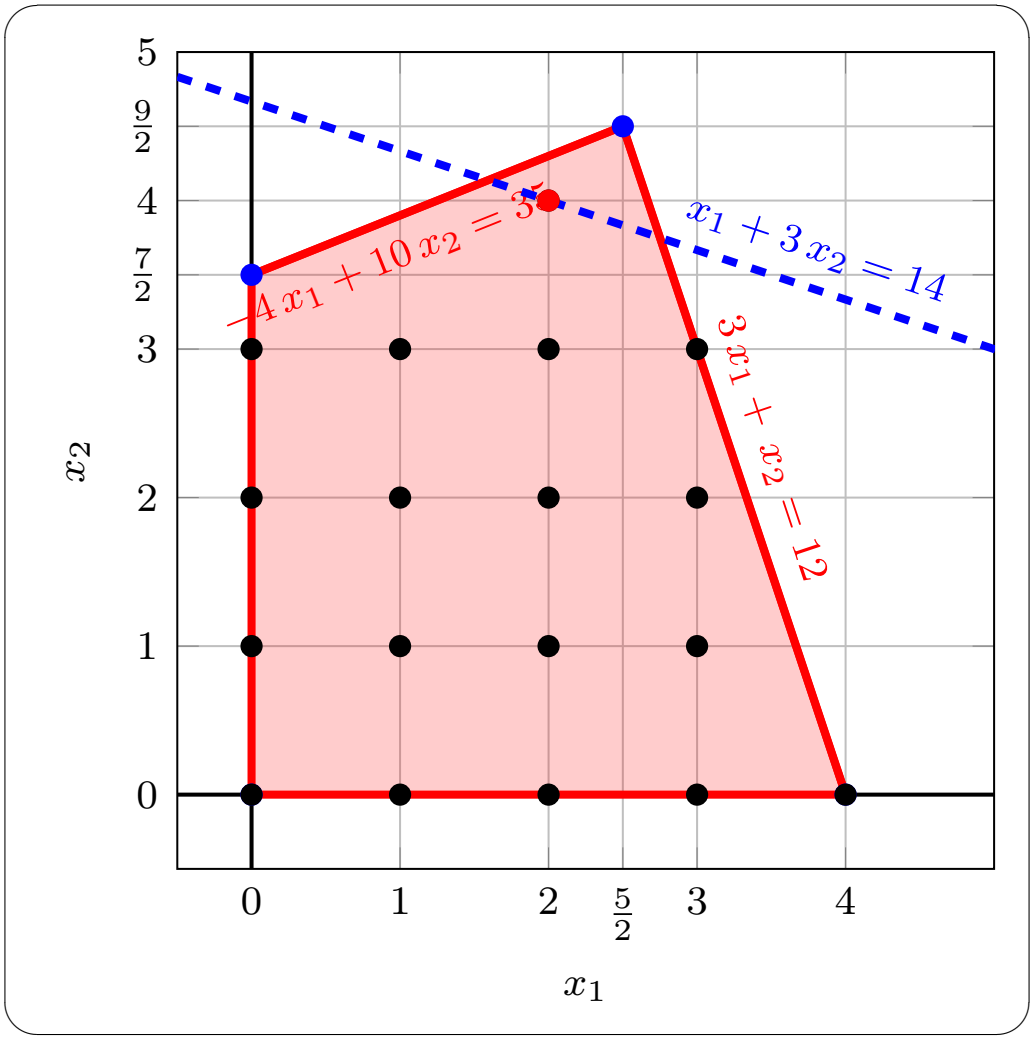
1.2 ILP problem *P1*

$$
\begin{array}{lrcrcl}
\max & x_1 & + & 8\ x_2 & & \\
\text{subject to} & 3\ x_1 & + & 6\ x_2 & \leq & 16 \\
& x_1, & & x_2 & \in & \mathbb{Z}_{\geq 0}
\end{array}
$$



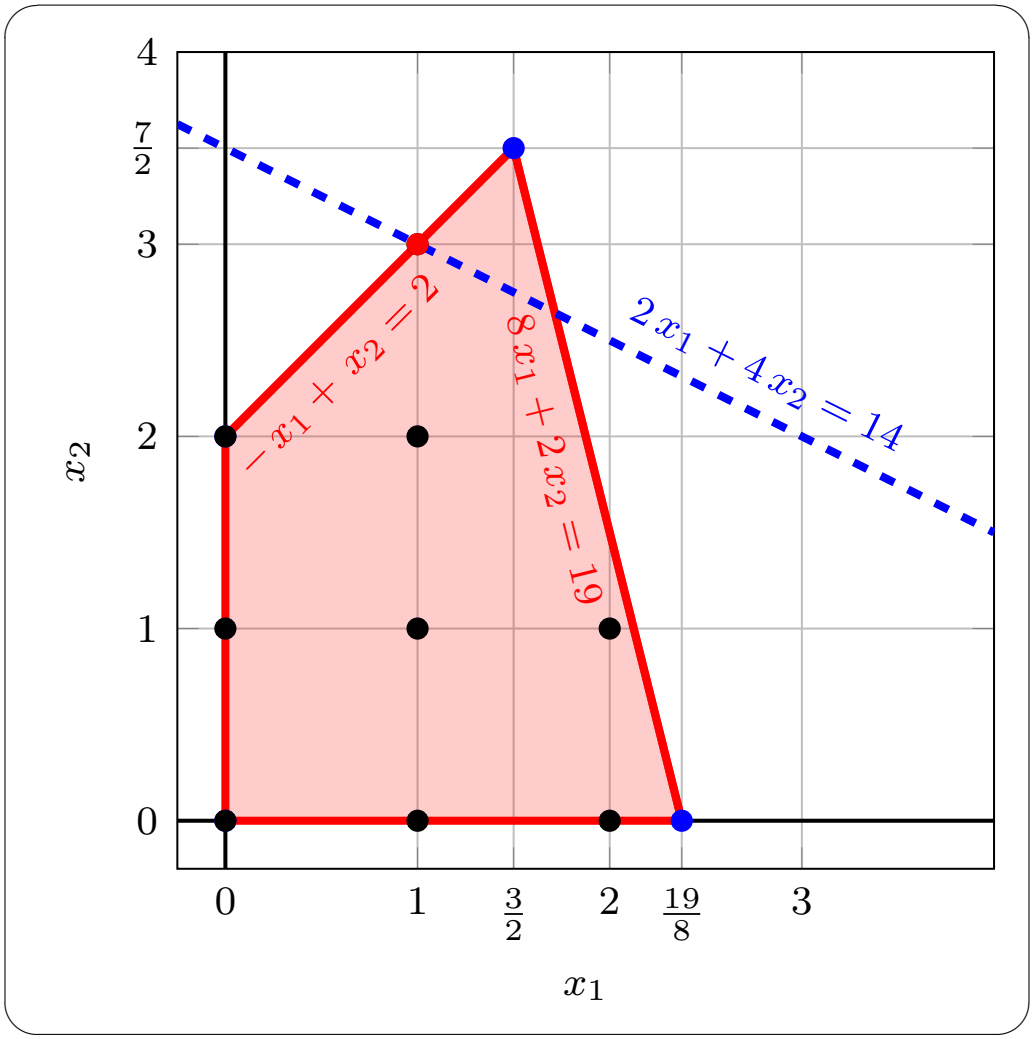
1.3 ILP problem *P2*

$$
\begin{array}{rrcrcrl}
\max & & x_1 & + & 3 & x_2 & \\
\text{subject to} & -4 & x_1 & + & 10 & x_2 & \leq 35 \\
& 3 & x_1 & + & & x_2 & \leq 12 \\
& & x_1, & & & x_2 & \in \mathbb{Z}_{\geq 0}
\end{array}
$$



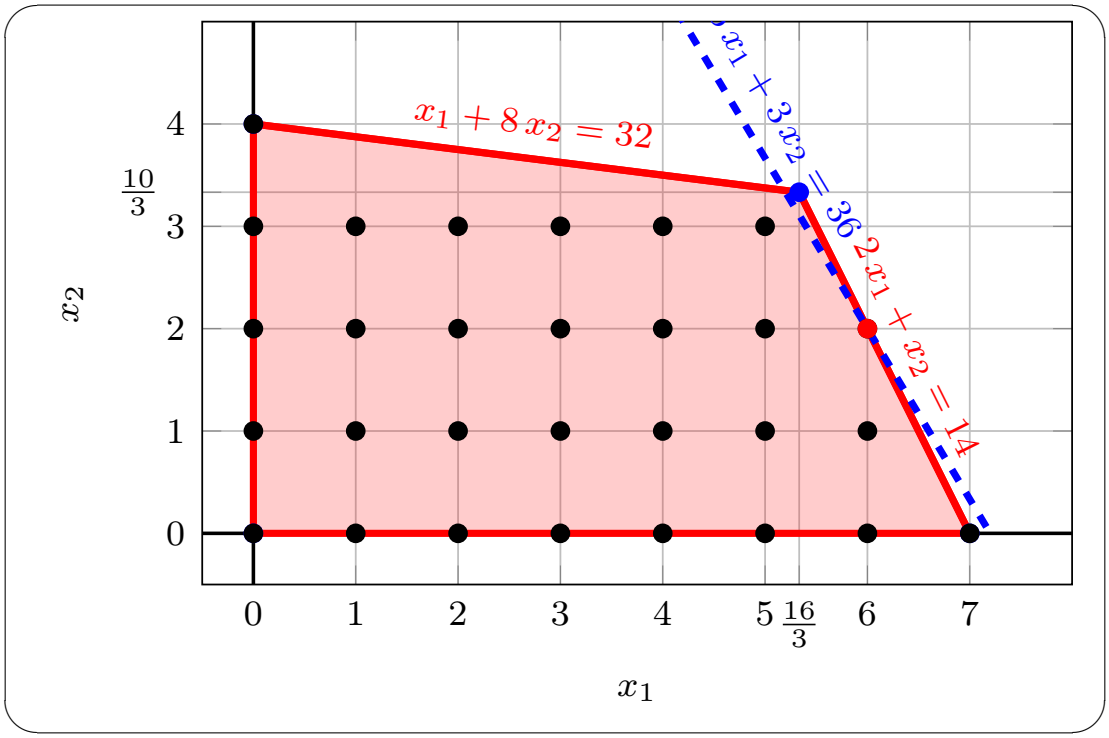
1.4 ILP problem $P3$

$$
\begin{array}{rrcrcrl}
\max & 2 & x_1 & + & 4 & x_2 & \\
\text{subject to} & -1 & x_1 & + & 1 & x_2 & \leq 2 \\
& 8 & x_1 & + & 2 & x_2 & \leq 19 \\
& & x_1, & & & x_2 & \in \mathbb{Z}_{\geq 0}
\end{array}
$$



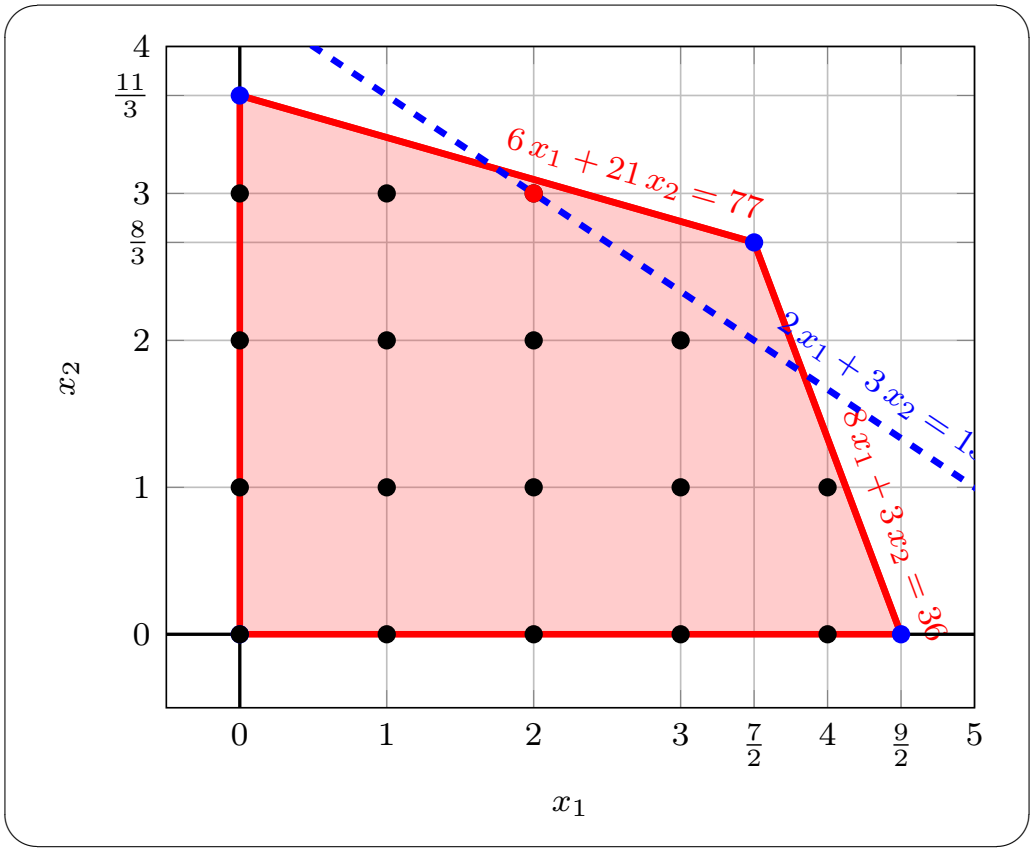
1.5 ILP problem *P4*

$$
\begin{array}{rrcrcl}
\max & 5\,x_1 & + & 3\,x_2 & & \\
\text{subject to} & 2\,x_1 & + & x_2 & \leq & 14 \\
& x_1 & + & 8\,x_2 & \leq & 32 \\
& x_1, & & x_2 & \in & \mathbb{Z}_{\geq 0}
\end{array}
$$



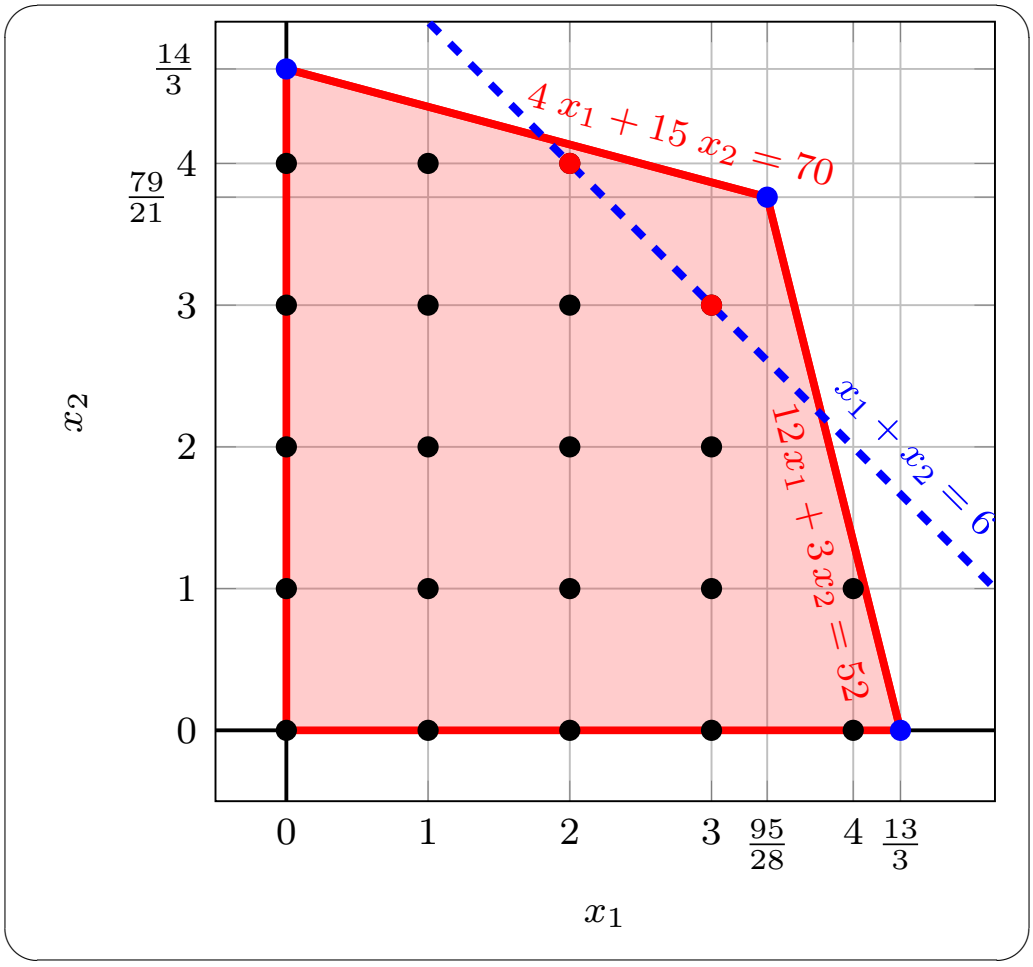
1.6 ILP problem $P5$

$$
\begin{array}{rrcrcl}
\max & 2\,x_1 & + & 3\,x_2 & & \\
\text{subject to} & 6\,x_1 & + & 21\,x_2 & \leq & 77 \\
& 8\,x_1 & + & 3\,x_2 & \leq & 36 \\
& x_1, & & x_2 & \in & \mathbb{Z}_{\geq 0}
\end{array}
$$



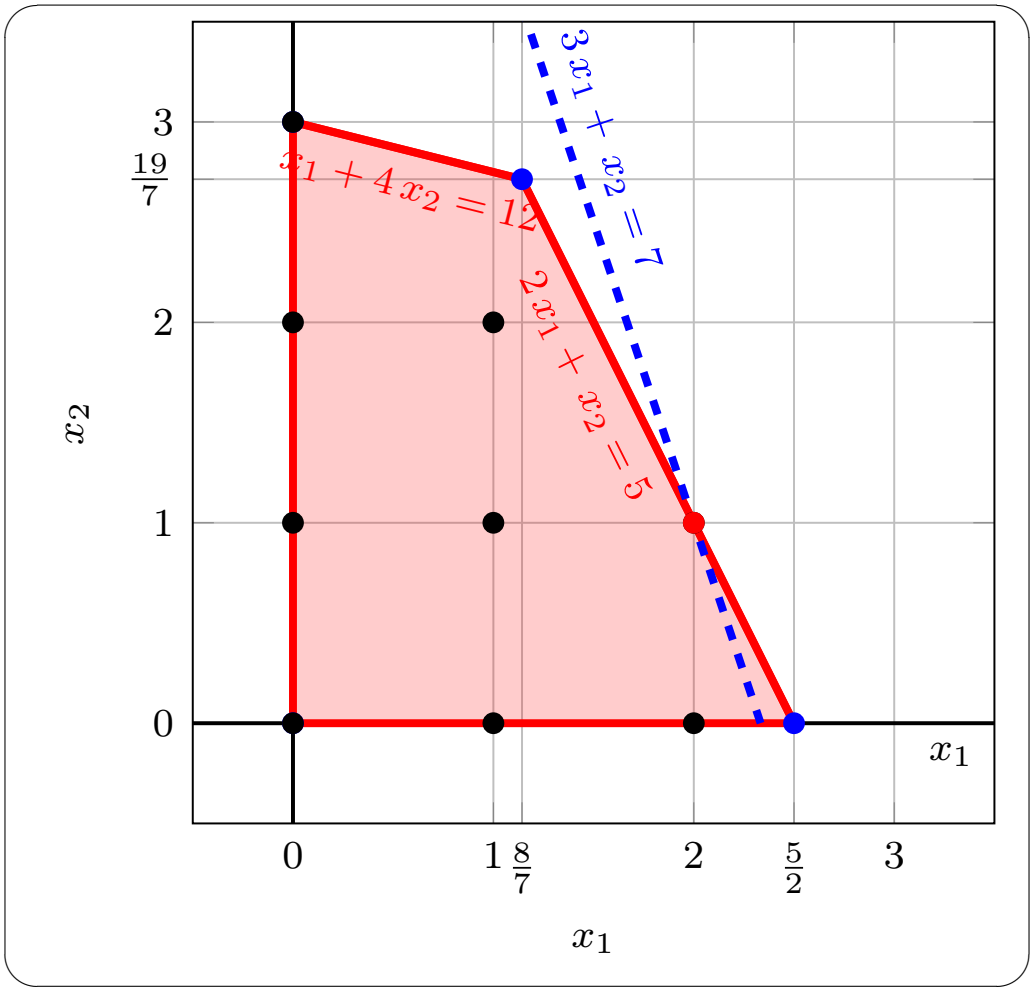
1.7 ILP problem *P6*

$$
\begin{array}{rrcrcrl}
\max & x_1 & + & & x_2 & & \\
\text{subject to} & 4\,x_1 & + & 15\,x_2 & \leq & 70 \\
& 12\,x_1 & + & 3\,x_2 & \leq & 52 \\
& x_1, & & & x_2 & \in & \mathbb{Z}_{\geq 0}
\end{array}
$$



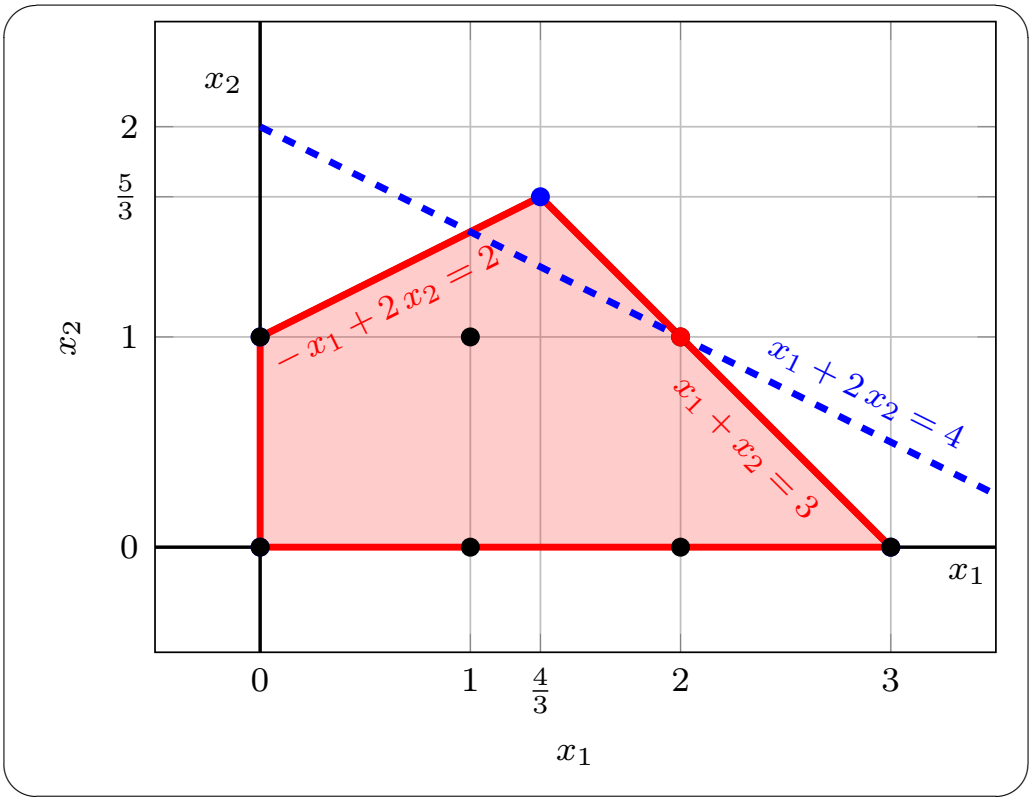
1.8 ILP problem *P7*

$$
\begin{array}{rrcrcrcr}
\max & 3 & x_1 & + & & x_2 & & \\
\text{subject to} & & x_1 & + & 4 & x_2 & \leq & 12 \\
& 2 & x_1 & + & & x_2 & \leq & 5 \\
& & x_1, & & & x_2 & \in & \mathbb{Z}_{\geq 0}
\end{array}
$$



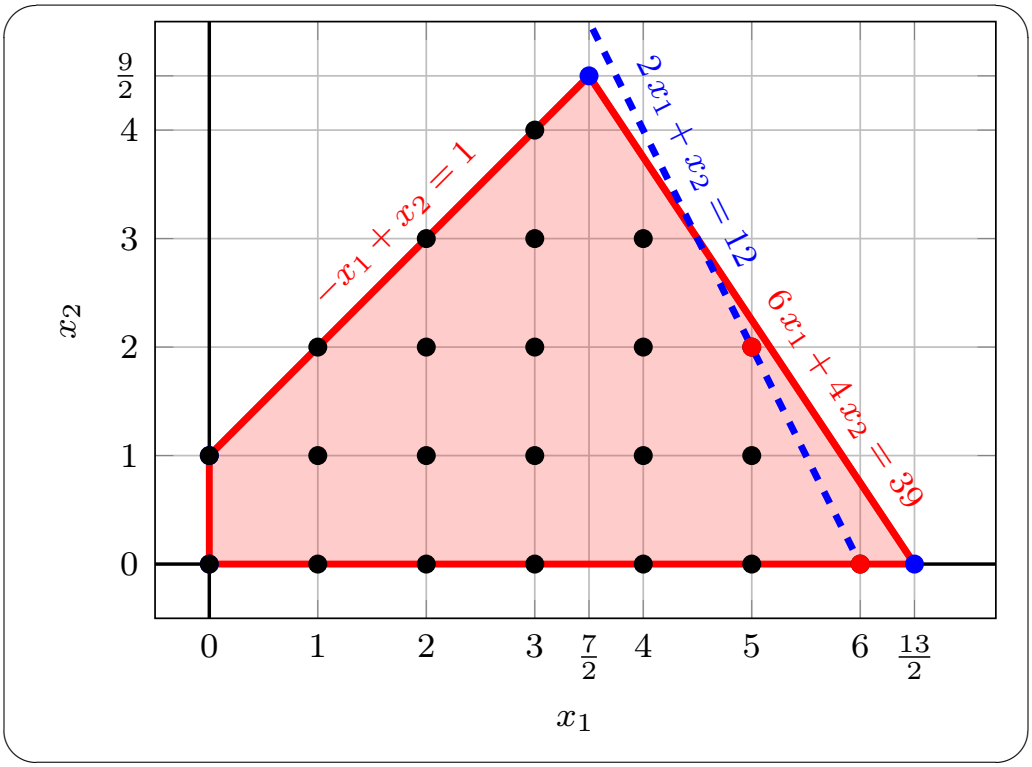
1.9 ILP problem *P8*

$$
\begin{array}{rrcrcl}
\max & x_1 & + & 2\ x_2 & & \\
\text{subject to} & -\ x_1 & + & 2\ x_2 & \leq & 2 \\
 & x_1 & + & x_2 & \leq & 3 \\
 & x_1, & & x_2 & \in & \mathbb{Z}_{\geq 0}
\end{array}
$$



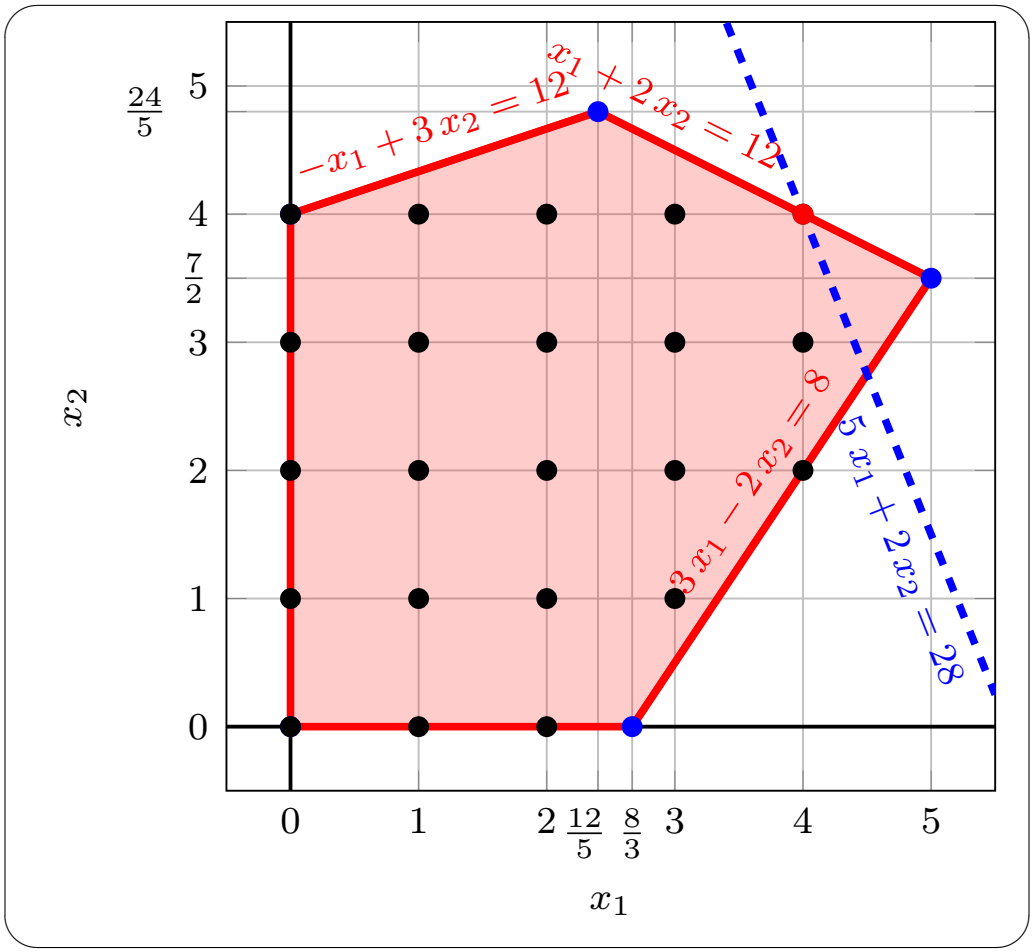
1.10 ILP problem *P9*

$$
\begin{array}{rrcrcrcr}
\max & 2 & x_1 & + & & x_2 & & \\
\text{subject to} & - & x_1 & + & & x_2 & \leq & 1 \\
& 6 & x_1 & + & 4 & x_2 & \leq & 39 \\
& & x_1, & & & x_2 & \in & \mathbb{Z}_{\geq 0}
\end{array}
$$



1.11 ILP problem *P*10

$$
\begin{array}{rrcrcrl}
\max & 5 & x_1 & + & 2 & x_2 & \\
\text{subject to} & - & x_1 & + & 3 & x_2 & \leq 12 \\
 & & x_1 & + & 2 & x_2 & \leq 12 \\
 & 3 & x_1 & - & 2 & x_2 & \leq 8 \\
 & & x_1, & & & x_2 & \in \mathbb{Z}_{\geq 0}
\end{array}
$$

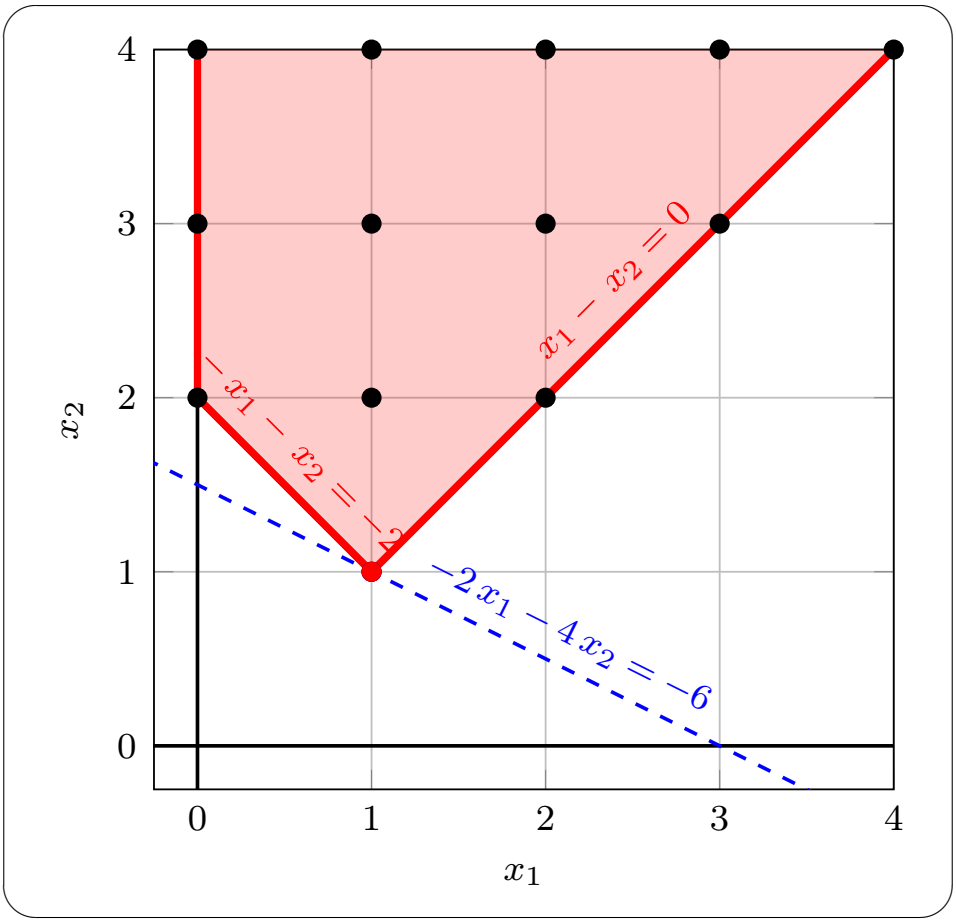


Chapter 2

ILP problems with two variables – Type B

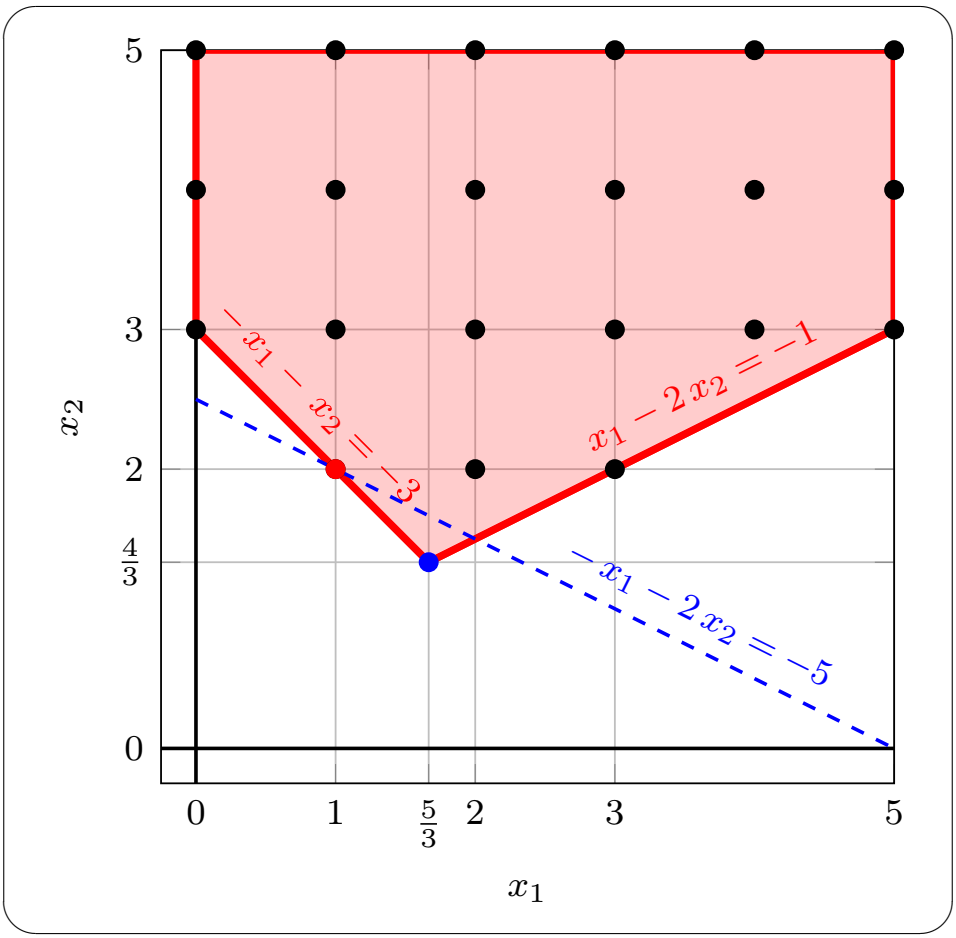
2.1 ILP problem *P*11

$$
\begin{array}{rrcrcr}
\max & -2\,x_1 & - & 4\,x_2 & & \\
\text{subject to} & -\,x_1 & - & x_2 & \leq & -2 \\
& x_1 & - & x_2 & \leq & 0 \\
& x_1, & & x_2 & \in & \mathbb{Z}_{\geq 0}
\end{array}
$$



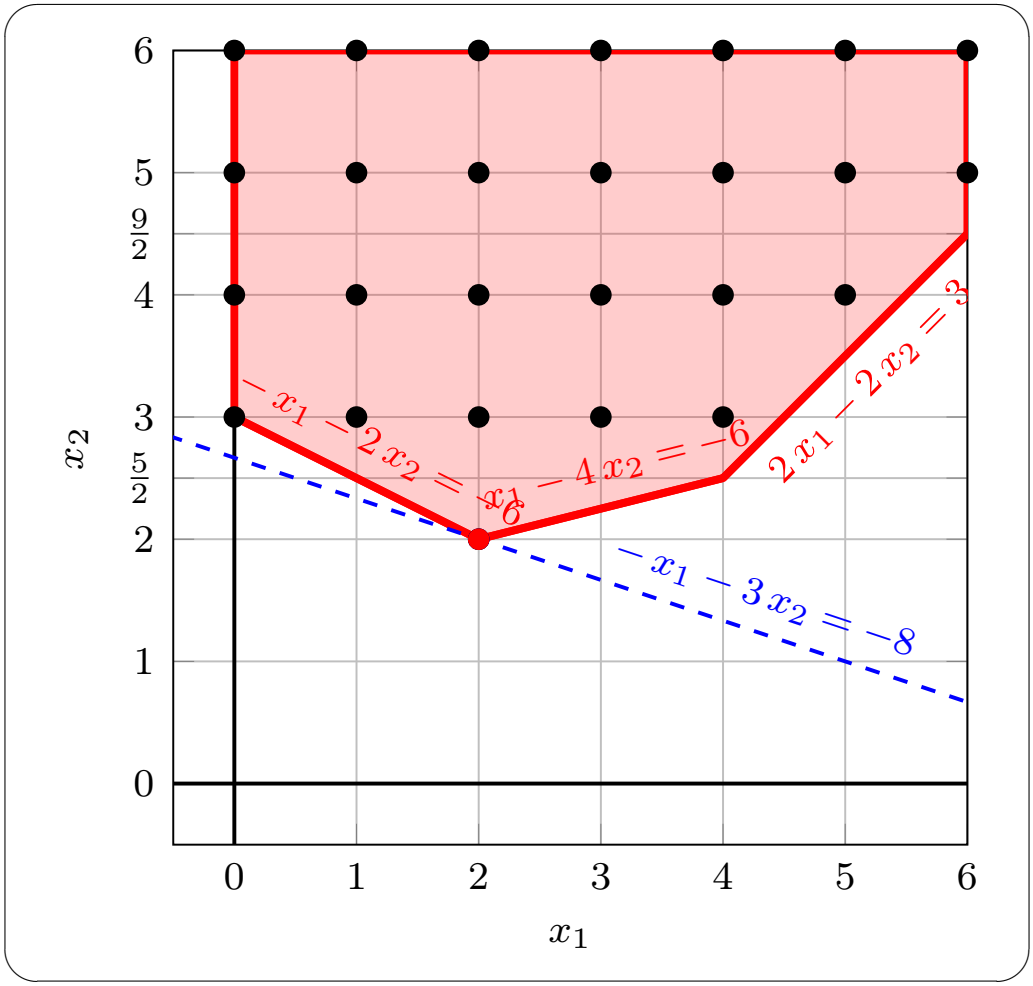
2.2 ILP Problem *P*12

$$
\begin{array}{rrrrrr}
\max & -x_1 & - & 2\,x_2 & & \\
\text{subject to} & -x_1 & - & x_2 & \leq & -3 \\
& x_1 & - & 2\,x_2 & \leq & -1 \\
& x_1, & & x_2 & \in & \mathbb{Z}_{\geq 0}
\end{array}
$$



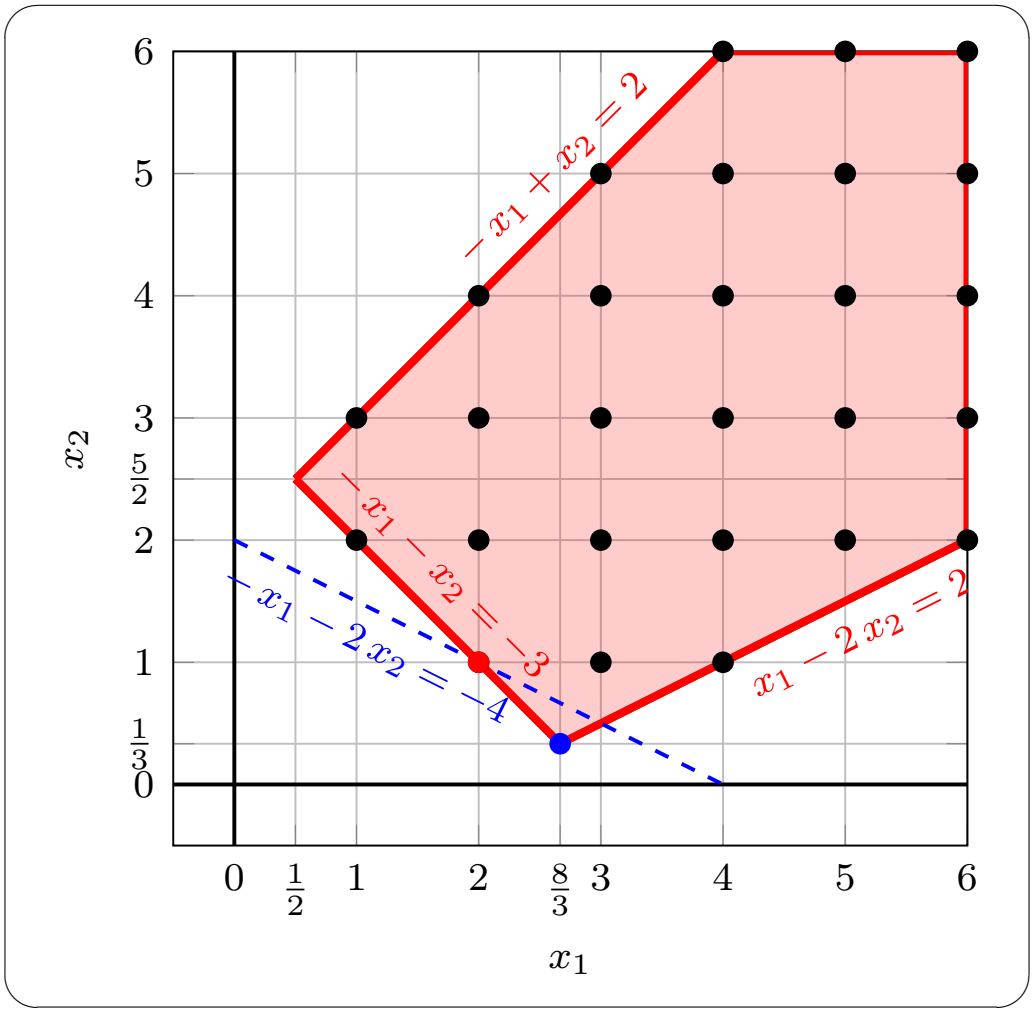
2.3 ILP problem *P*13

$$
\begin{array}{rrrrrrr}
\max & - & x_1 & - & 3\ x_2 & & \\
\text{subject to} & - & x_1 & - & 2\ x_2 & \leq & -6 \\
 & & x_1 & - & 4\ x_2 & \leq & -6 \\
 & 2\ & x_1 & - & 2\ x_2 & \leq & 3 \\
 & & x_1, & & x_2 & \in & \mathbb{Z}_{\geq 0}
\end{array}
$$



2.4 ILP Problem *P*14

$$
\begin{array}{rrrrrrrr}
\max & - & x_1 & - & 2 & x_2 & & \\
\text{subject to} & - & x_1 & + & & x_2 & \leq & 2 \\
& & x_1 & - & 2 & x_2 & \leq & 2 \\
& - & x_1 & - & & x_2 & \leq & -3 \\
& & x_1, & & & x_2 & \in & \mathbb{Z}_{\geq 0}
\end{array}
$$

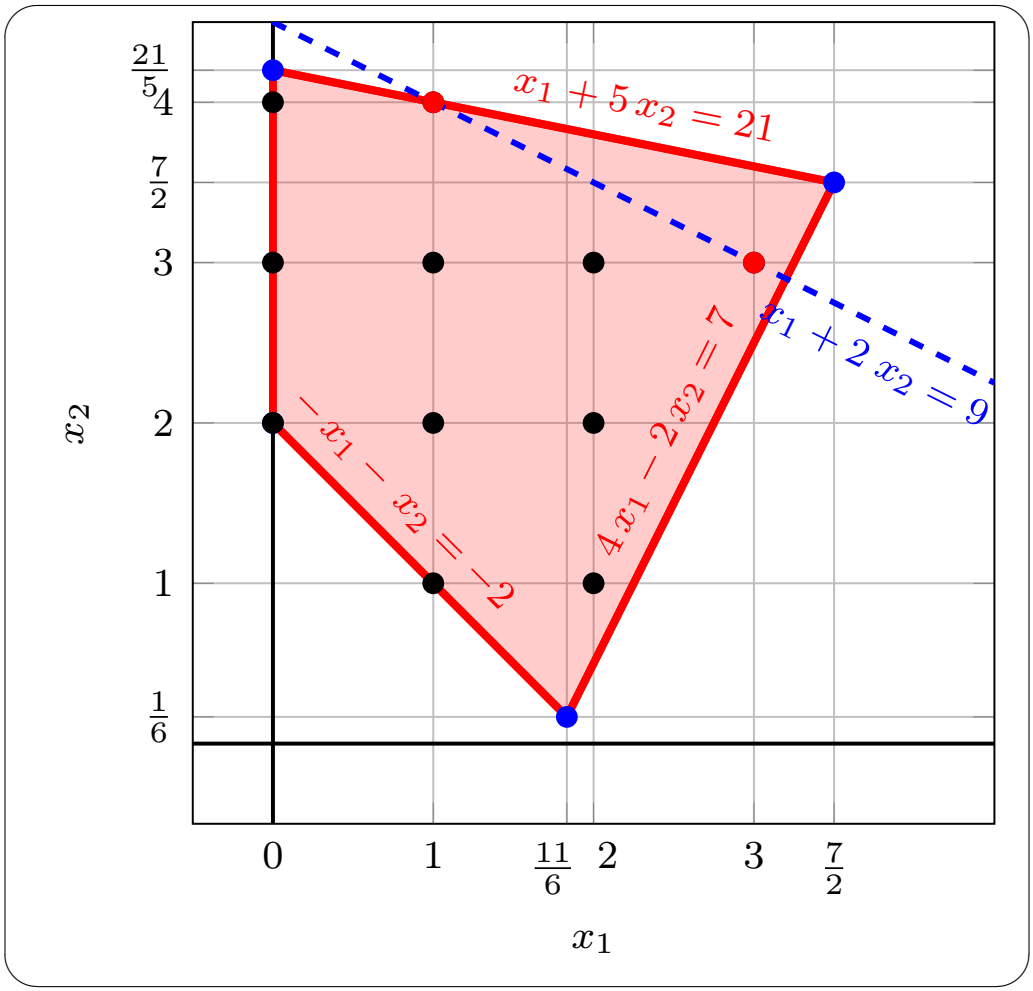


Chapter 3

ILP problems with two variables – Type C

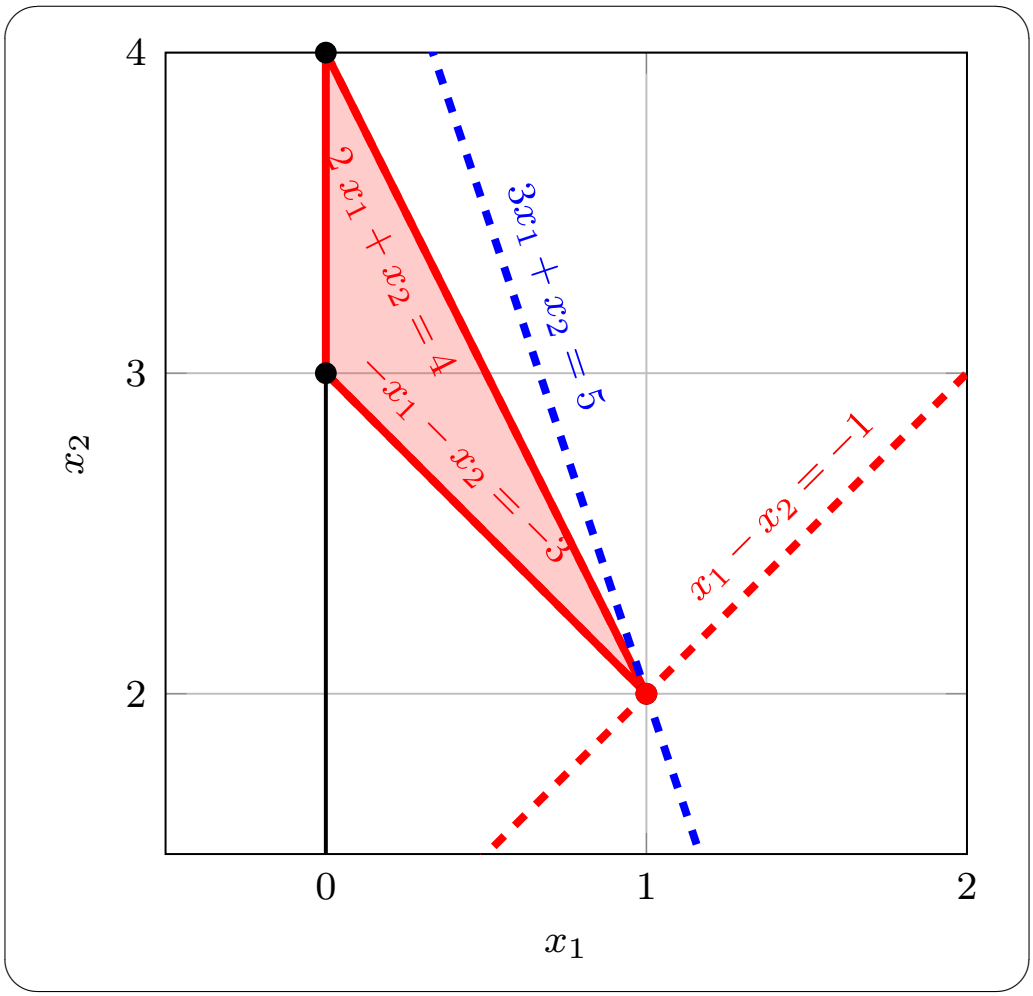
3.1 ILP problem *P*21

$$
\begin{array}{rrcrcrr}
\max & x_1 & + & 2 & x_2 & & \\
\text{subject to} & -x_1 & - & & x_2 & \leq & -2 \\
& 4\,x_1 & - & 2 & x_2 & \leq & 7 \\
& x_1 & + & 5 & x_2 & \leq & 21 \\
& x_1, & & & x_2 & \in & \mathbb{Z}_{\geq 0}
\end{array}
$$



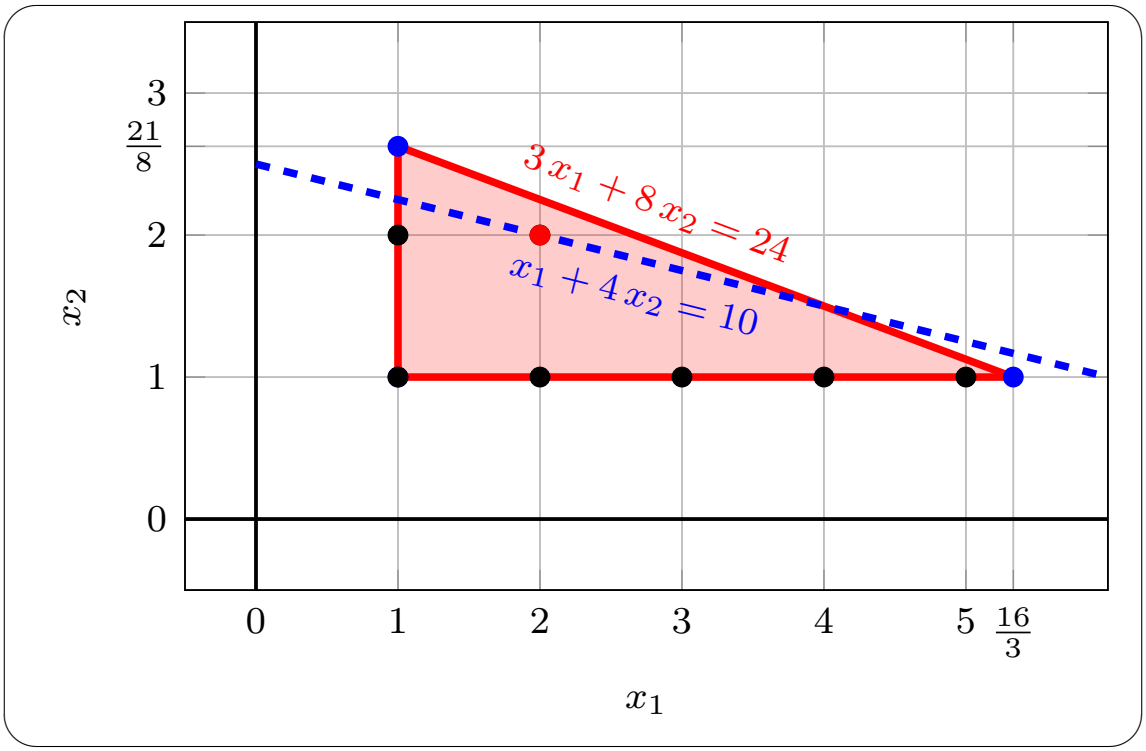
3.2 ILP problem *P22*

$$
\begin{array}{rrcrcr}
\max & 3\,x_1 & + & x_2 & & \\
\text{subject to} & x_1 & - & x_2 & \leq & -1 \\
& -\,x_1 & - & x_2 & \leq & -3 \\
& 2\,x_1 & + & x_2 & \leq & 4 \\
& x_1, & & x_2 & \in & \mathbb{Z}_{\geq 0}
\end{array}
$$



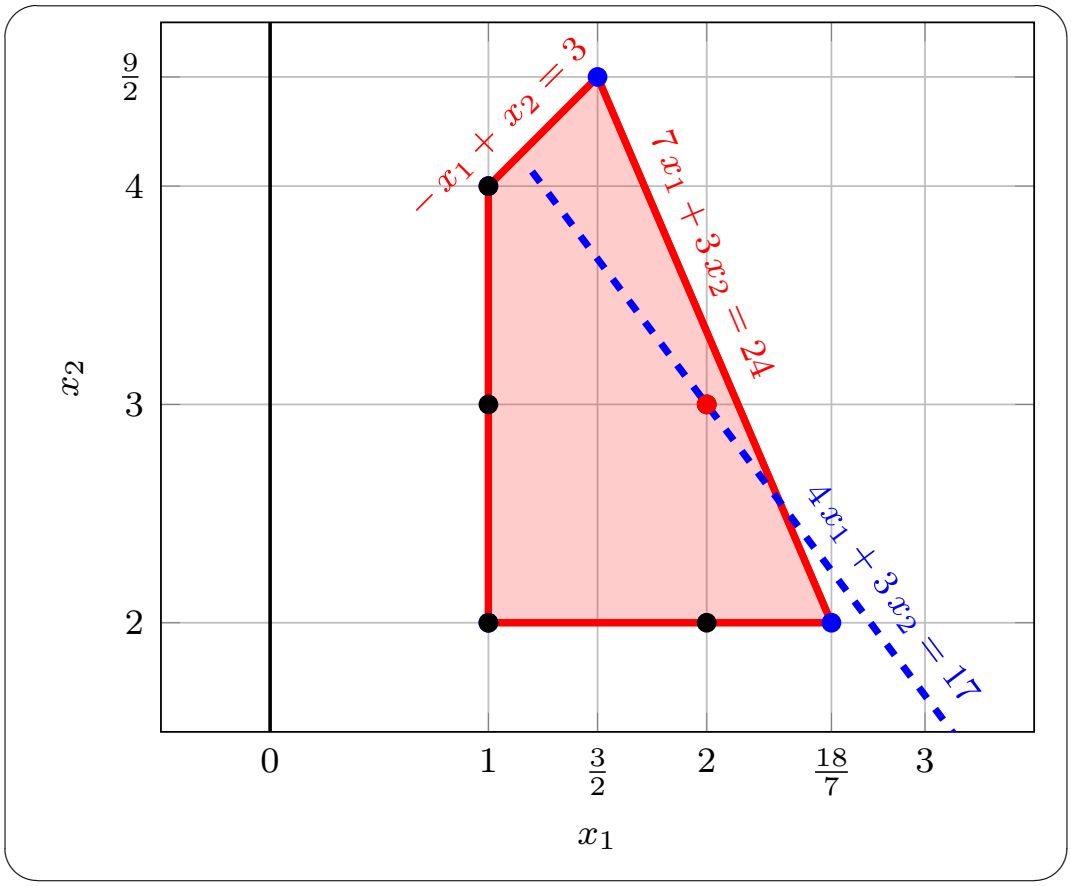
3.3 ILP problem *P23*

$$
\begin{array}{lrcrcrl}
\max & & x_1 & + & 4 & x_2 & \\
\text{subject to} & 3 & x_1 & + & 8 & x_2 & \leq & 24 \\
 & & x_1 & & & & \geq & 1 \\
 & & & & & x_2 & \geq & 1 \\
 & & x_1, & & & x_2 & \in & \mathbb{Z}_{\geq 0}
\end{array}
$$



3.4 ILP problem *P24*

$$
\begin{array}{rrcrcrl}
\max & 4\,x_1 & + & 3\,x_2 & & & \\
\text{subject to} & 7\,x_1 & + & 3\,x_2 & \leq & 24 \\
& -\,x_1 & + & x_2 & \leq & 3 \\
& x_1 & & & \geq & 1 \\
& & & x_2 & \geq & 2 \\
& x_1, & & x_2 & \in & \mathbb{Z}_{\geq 0}
\end{array}
$$

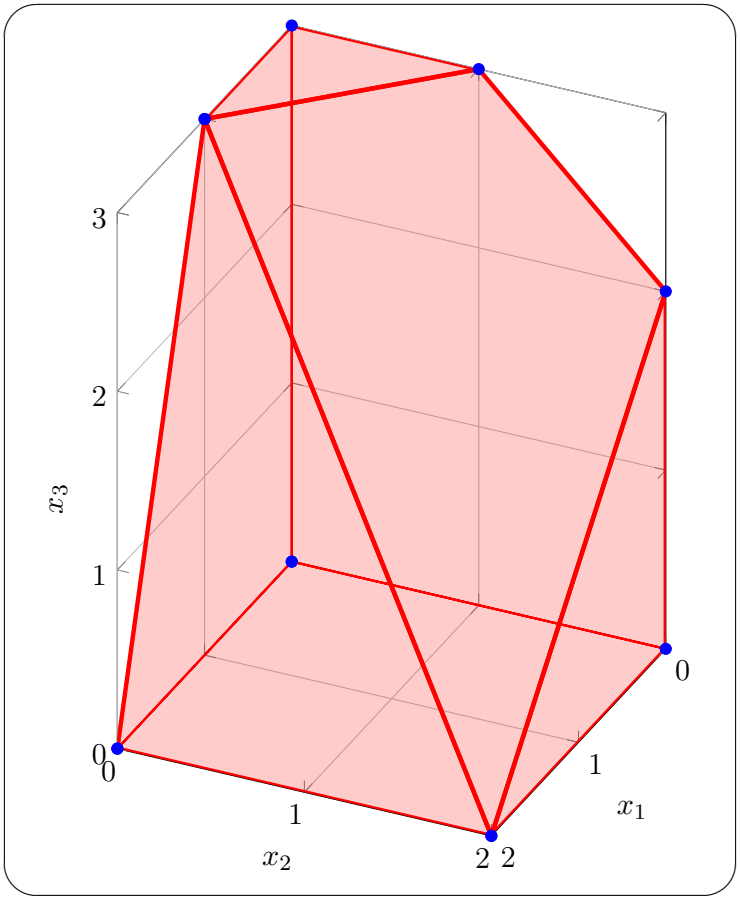


Chapter 4

ILP problems with three variables – Type A

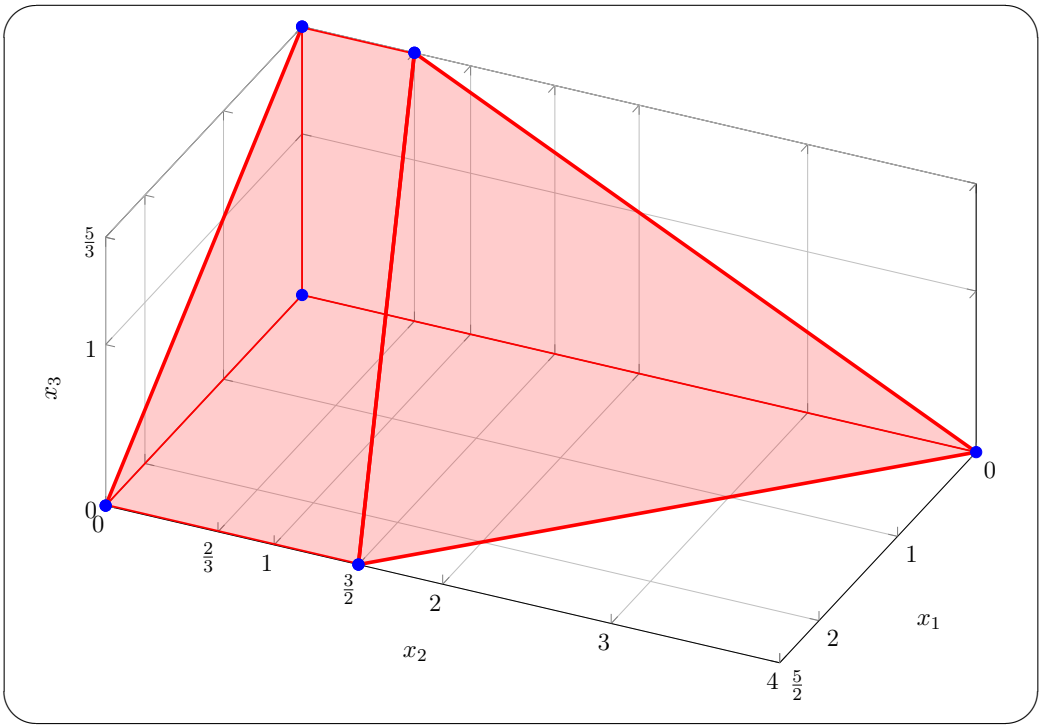
4.1 ILP problem *P*41

$$
\begin{array}{rrcrcrcl}
\max & 2\,x_1 & + & x_2 & + & 3\,x_3 & & \\
\text{subject to} & x_1 & + & x_2 & + & x_3 & \leq & 4 \\
 & & & x_2 & & & \leq & 2 \\
 & & & & & x_3 & \leq & 3 \\
 & 3\,x_1 & & & + & x_3 & \leq & 6 \\
 & x_1, & & x_2, & & x_3 & \in & \mathbb{Z}_{\geq 0}
\end{array}
$$



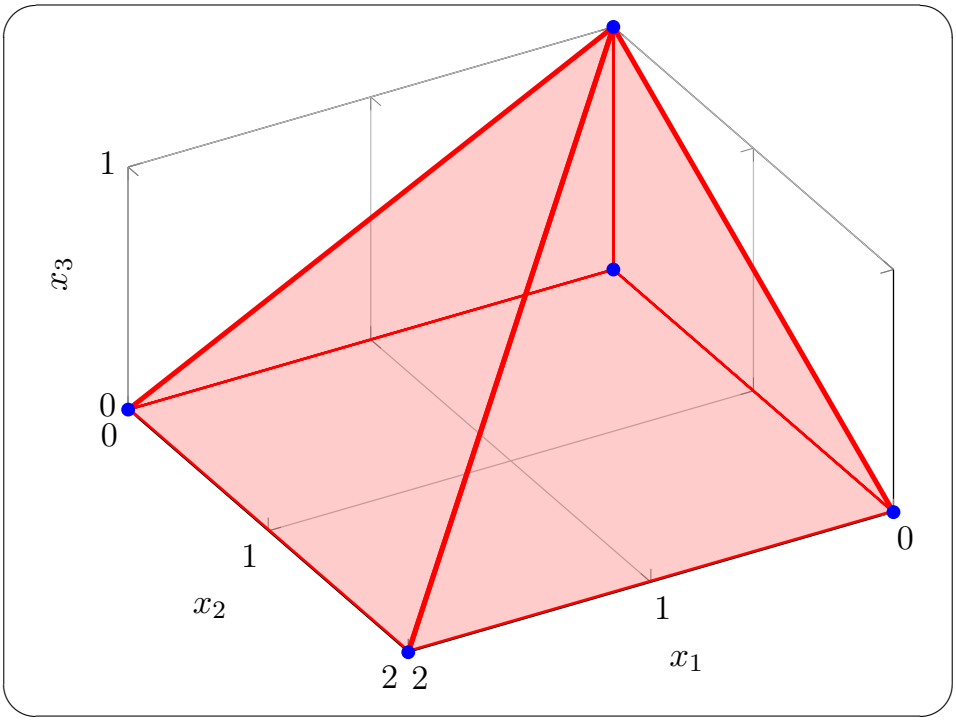
4.2 ILP problem *P42*

$$
\begin{array}{rrcrcrcrcl}
\max & 3 & x_1 & + & 2 & x_2 & + & 4 & x_3 & \\
\text{subject to} & & x_1 & + & & x_2 & + & 2 & x_3 & \leq & 4 \\
& 2 & x_1 & & & & + & 3 & x_3 & \leq & 5 \\
& 2 & x_1 & + & & x_2 & + & 3 & x_3 & \leq & 7 \\
& & x_1, & & & x_2, & & & x_3 & \in & \mathbb{Z}_{\geq 0}
\end{array}
$$



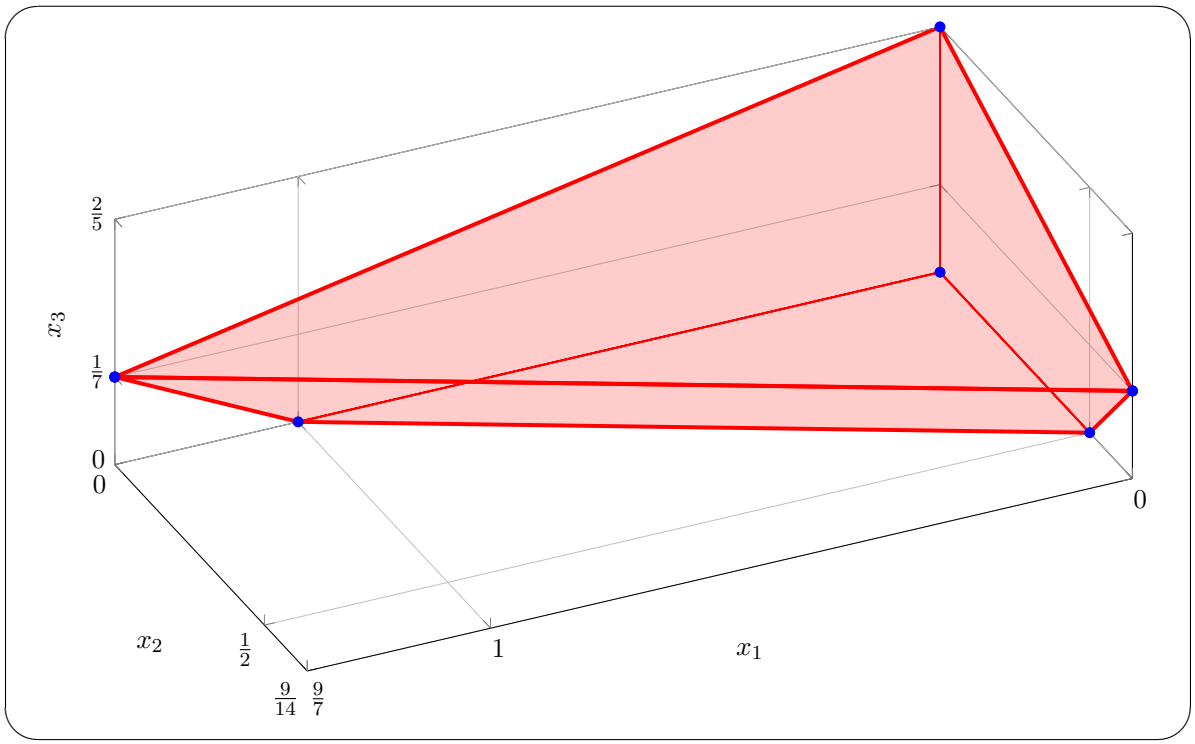
4.3 ILP problem *P43*

$$
\begin{array}{rrcrcrcl}
\max & x_1 & + & 2\ x_2 & + & 7\ x_3 & & \\
\text{subject to} & x_1 & & & + & 2\ x_3 & \leq & 2 \\
& & & x_2 & + & 2\ x_3 & \leq & 2 \\
& x_1, & & x_2, & & x_3 & \in & \mathbb{Z}_{\geq 0}
\end{array}
$$



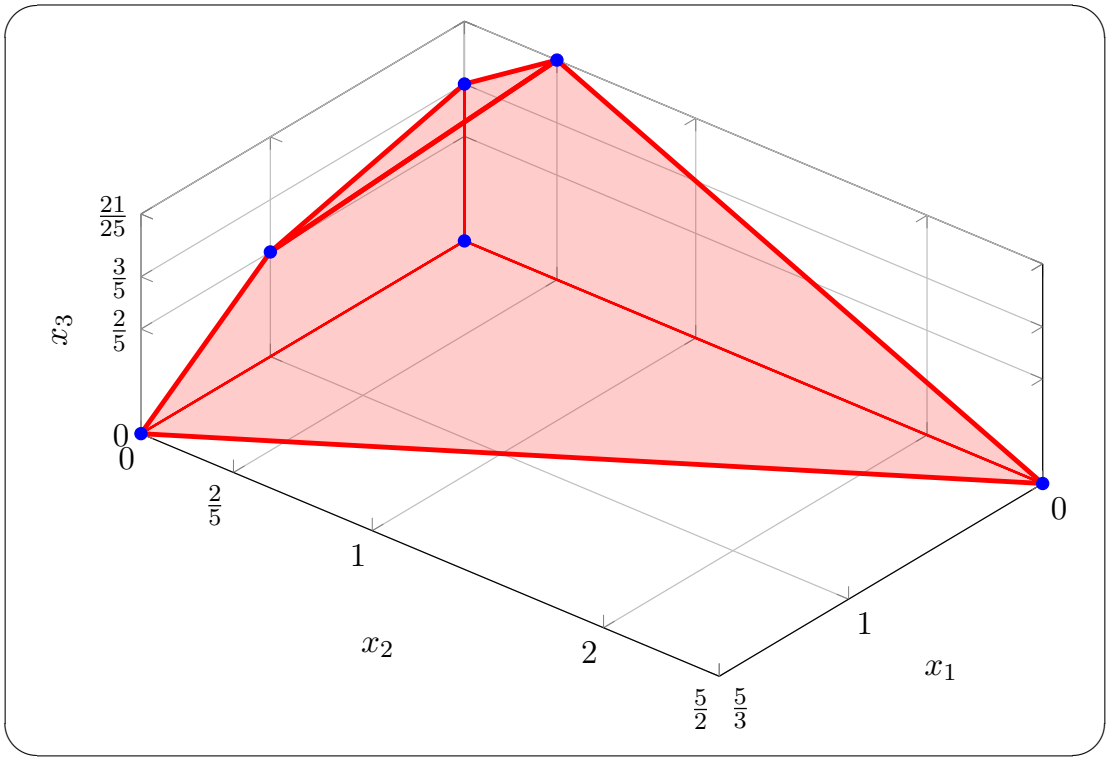
4.4 ILP problem *P44*

$$
\begin{array}{rrcrcrcrl}
\max & x_1 & + & 2\,x_2 & + & 3\,x_3 & & & \\
\text{subject to} & x_1 & + & 2\,x_2 & - & 2\,x_3 & \leq & 1 \\
2\,x_1 & - & & x_2 & - & 3\,x_3 & \leq & 4 \\
x_1 & + & 2\,x_2 & + & 5\,x_3 & & \leq & 2 \\
x_1, & & & x_2, & & x_3 & \in & \mathbb{Z}_{\geq 0}
\end{array}
$$



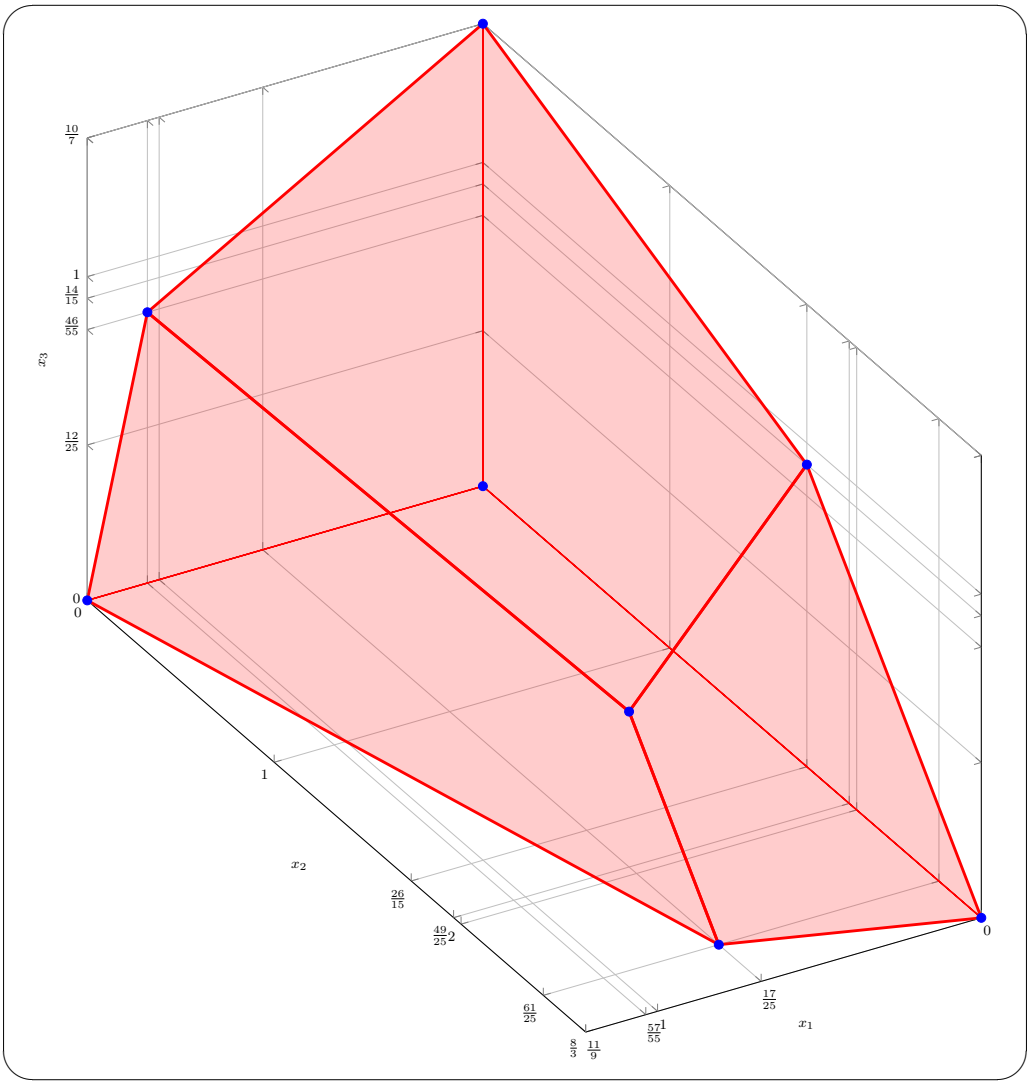
4.5 ILP problem *P45*

$$
\begin{array}{rrcrcrcrl}
\max & 4 & x_1 & - & 13\,x_2 & + & 7 & x_3 & \\
\text{subject to} & 3\,x_1 & + & 2 & x_2 & + & 5 & x_3 & \leq & 5 \\
& x_1 & - & 3 & x_2 & + & 5 & x_3 & \leq & 3 \\
& x_1, & & & x_2, & & & x_3 & \in & \mathbb{Z}_{\geq 0}
\end{array}
$$



4.6 ILP Problem *P46*

$$
\begin{array}{rrcrcrcl}
\max & 5\ x_1 & + & 6\ x_2 & + & 10\ x_3 & & \\
\text{subject to} & 9\ x_1 & + & 2\ x_2 & + & 2\ x_3 & \leq & 11 \\
& 2\ x_1 & + & 6\ x_2 & + & 6\ x_3 & \leq & 16 \\
& 4\ x_1 & + & 2\ x_2 & + & 7\ x_3 & \leq & 10 \\
& x_1, & & x_2, & & x_3 & \in & \mathbb{Z}_{\geq 0}
\end{array}
$$

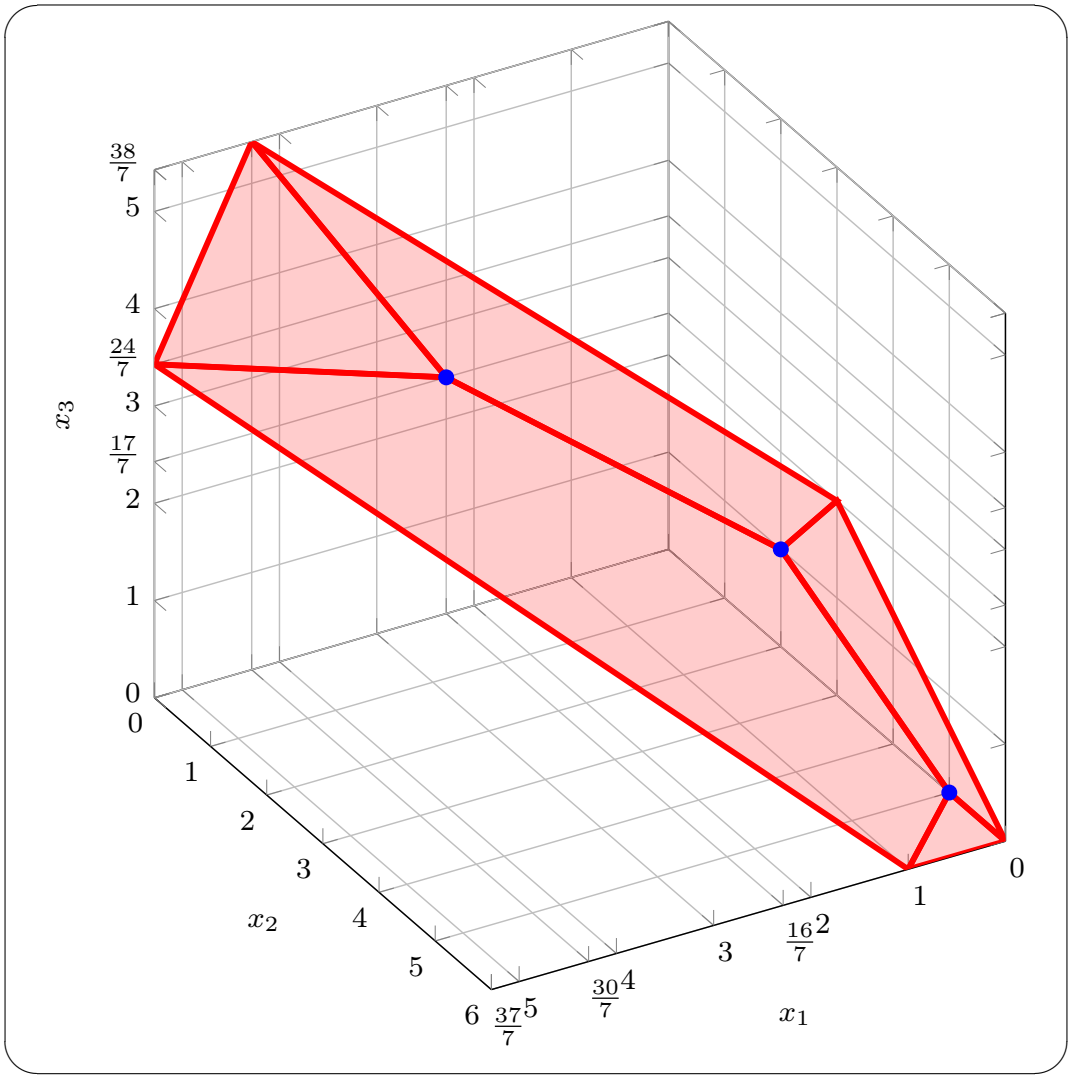


Chapter 5

ILP problems with three variables – Type B

5.1 ILP problem $P51$

$$
\begin{array}{rrrrrrrrrrr}
\max & - & 2\,x_1 & - & x_2 & & & & & & \\
\text{subject to} & & x_1 & - & x_2 & - & 3 & x_3 & \leq & -5 \\
& - & 3 & x_1 & - & 2 & x_2 & + & 2 & x_3 & \leq & -2 \\
& & & x_1, & & & x_2, & & & x_3 & \in & \mathbb{Z}_{\geq 0}
\end{array}
$$



Chapter 6

ILP problems with three variables – Type C

6.1 ILP problem $P61$

Chapter 7

ILP problems with several variables

7.1 ILP problem *P*71

$$
\begin{array}{rrcrcrcrcrcr}
\max & 4 & x_1 & + & x_2 & + & 5 & x_3 & + & 3 & x_4 & \\
\text{subject to} & & x_1 & - & x_2 & - & & x_3 & + & 3 & x_4 & \leq & 1 \\
& 5 & x_1 & + & x_2 & + & 3 & x_3 & + & 8 & x_4 & \leq & 55 \\
& - & x_1 & + & 2\ x_2 & + & 3 & x_3 & - & 5 & x_4 & \leq & 3 \\
& & x_1, & & x_2, & & & x_3, & & & x_4 & \in & \mathbb{Z}_{\geq 0}
\end{array}
$$

7.2 ILP problem *P72*

$$
\begin{array}{rrrrrrrrrrrrrrrr}
\max & 7 & x_1 & + & 6 & x_2 & + & 5 & x_3 & - & 2 & x_4 & + & 3 & x_5 & \\
\text{subject to} & & x_1 & + & 3 & x_2 & + & 5 & x_3 & - & 2 & x_4 & + & 2 & x_5 & \leq & 4 \\
& 4 & x_1 & + & 2 & x_2 & - & 2 & x_3 & + & & x_4 & + & & x_5 & \leq & 3 \\
& 2 & x_1 & + & 4 & x_2 & + & 4 & x_3 & - & 2 & x_4 & + & 5 & x_5 & \leq & 5 \\
& 3 & x_1 & + & & x_2 & + & 2 & x_3 & - & & x_4 & - & 2 & x_5 & \leq & 1 \\
& & x_1, & & & x_2, & & & x_3, & & & x_4, & & & x_5 & \in & \mathbb{Z}_{\geq 0}
\end{array}
$$